

Sonification – aiming for the blind

CVL Master's Thesis Project Proposal



This project aims for tools for the blind, e.g. to help them do sports. The images above show the blind Olympic cross-country skier Brian McKeever with a guide, and a blind football game, where the ball emits sounds to tell the players where it is. We look for one or two students interested in developing algorithms for sonification of direction sensed by camera input, with the ultimate goal to increase the independence of blind people exercising sports.

We aim for a portable system that can be carried, but we will primarily explore and analyze the principles, and demonstrate performance in a lab environment (potentially using stationary equipment, if that is easier).

The project has many aspects and there is room for the student(s) to pick and choose what seems most interesting and promising.

Aspects of interest

1. Sonification calibration of directional perception for an individual. This can be done either by literature study, or by recordings from a small microphone situated in the ear of the user.
2. Create sonification of directional sensing of single visual events. In addition to auditory science papers there is also knowledge in the music industry about how to locate instruments e.g. in a symphony orchestra.
3. Experiment on sonifications of visually aided tasks. These may be e.g. to catch a balloon blindfolded.
4. Sonification of more complex, but still stylized scenes (auditory virtual reality). E.g. containing two persons and one ball in a room, slalom-walking between large foam-poles etc. How would you sonify a ball? a person? a foam-pole?
5. Collecting input from blind individuals for feedback.

Requirements

240hp, with courses in image and signal processing. Prior experience with raspberry Pi is an advantage.

Project Start

Open offer, earliest start is in January 2026.

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